I am going to collect health, nutrition, and physical activity measurements from the NHANES dataset, which is publicly available and includes body size data. My project will combine data from the 2015-2016 survey cycle with the 2017-2018 survey cycle to build a sufficiently large dataset. The four types of data I will utilize to perform my analysis include: DEMO (demographic data), DR1TOT (total daily nutrient intakes per individual), PAQ (physical activity variables), and BMX (individual body size data). To link these datasets, I will use the SEQN variable from the four datasets within each survey cycle, respectively, and then join within each survey cycle.

The evaluation of the NHANES database, which lists individual-level information on health, nutrition, and physical activity measures (which is publicly available), should use the body size data available in the NHANES database. I will use a combination of two survey cycles, 2015-2016 and 2017-2018, for developing my dataset (i.e., NHANES 2015-2016 and 2017-2018), which will be used in this analysis. The four types of data I will use in conducting my analysis include: DEMO (demographic data), DR1TOT (total amounts of nutrients consumed per person), PAQ (physical activity measures), and BMX (body size measurement). The method of combining these three data sets will be through the use of the SEQN variable from each data set within the survey cycle (for all three data sets in each survey cycle) and joining them within each of the two survey cycles.

I will complete the data preparation stage by selecting variables required for analysis, merging NHANES file data, correcting erroneous/missing values, renaming the variables for better clarity, and deriving new activity-related variables. I will then describe the data using descriptive statistics, visualizations, and a t-test to examine how differences in BMI vary by activity group. The primary statistical model will be linear regression, and I will examine the applicability of Ridge regression to determine whether there are collinearities between multiple predictors from the same dietary group.

This dataset will be suitable for the project because I can easily find a public, large enough dataset that connects directly to calorie tracking, macros, physical activity, and weight loss. However, one limitation of the dataset is that BMI does not distinguish between fat and muscle mass, so there are some NHANES variables that are self-reported. I plan to use Python in Google Colab to perform analysis and use Power BI so I can show you a quick dashboard with results from the analysis.